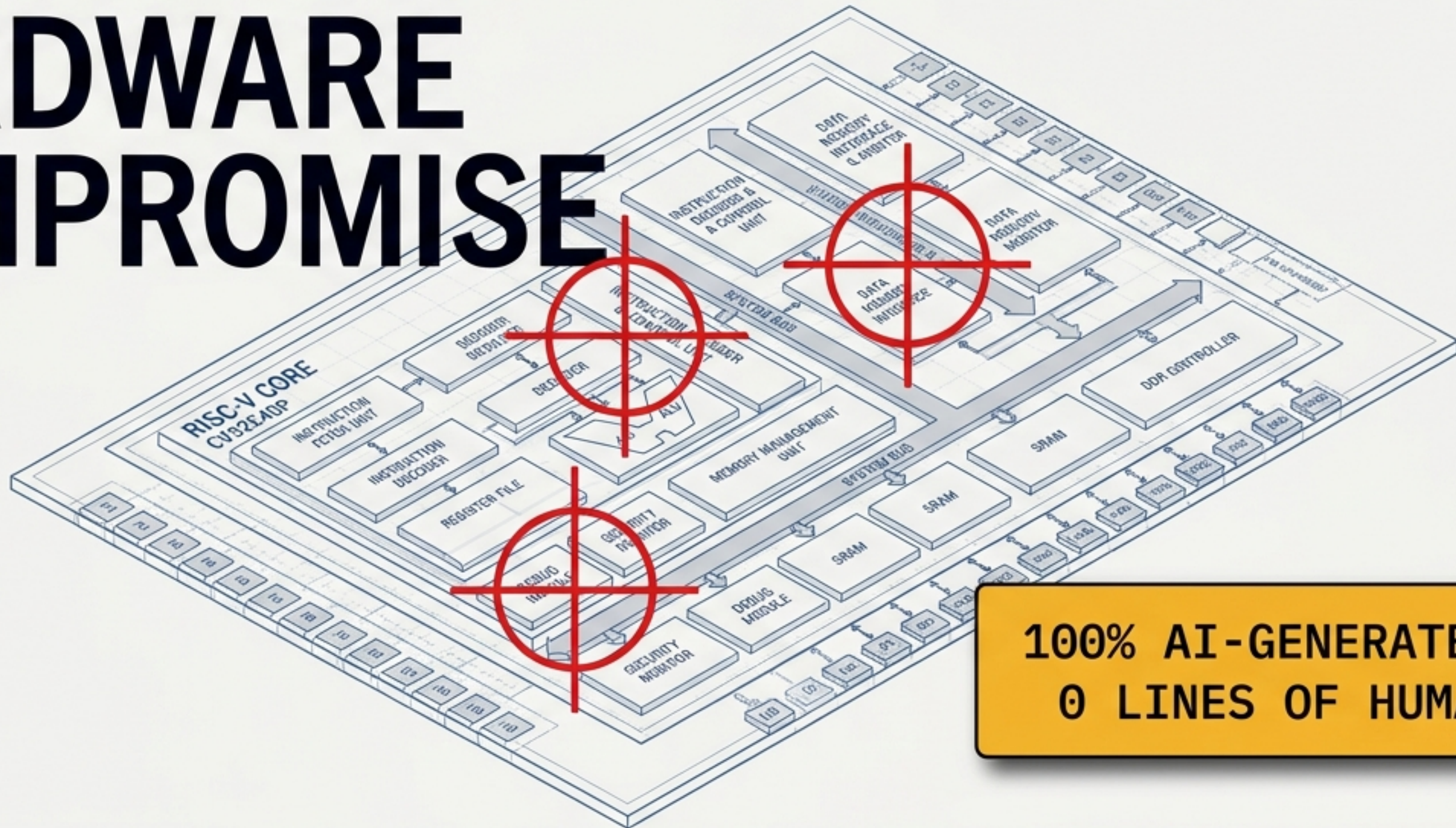


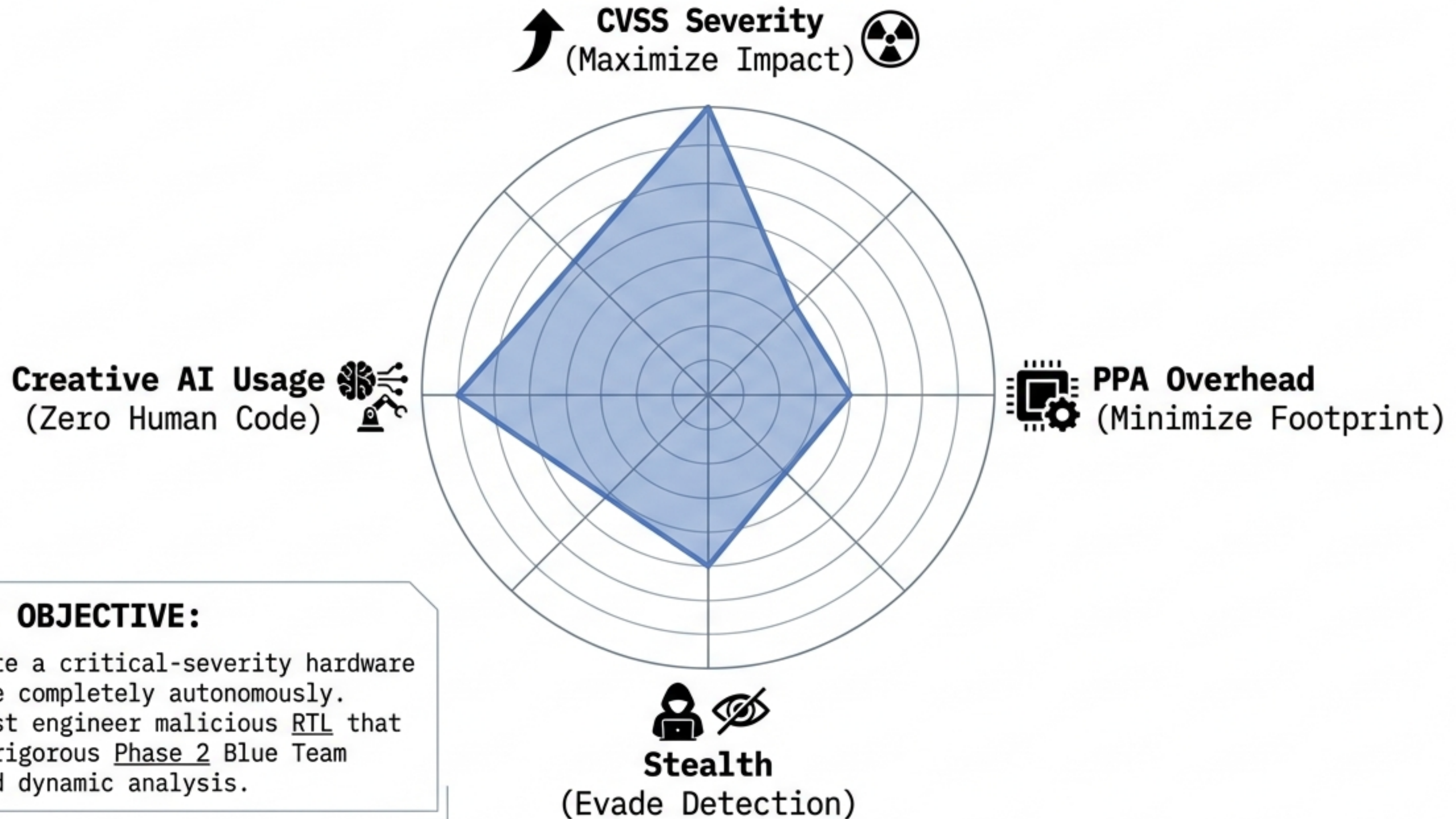
AUTONOMOUS HARDWARE COMPROMISE



100% AI-GENERATED RTL |
0 LINES OF HUMAN CODE

PROJECT REPORT: RED TEAM PHASE — CV32E40P RISC-V HARDWARE TROJAN INSERTION.
IEEE HOST 2026 AI HARDWARE ATTACK (AHA!) CHALLENGE.

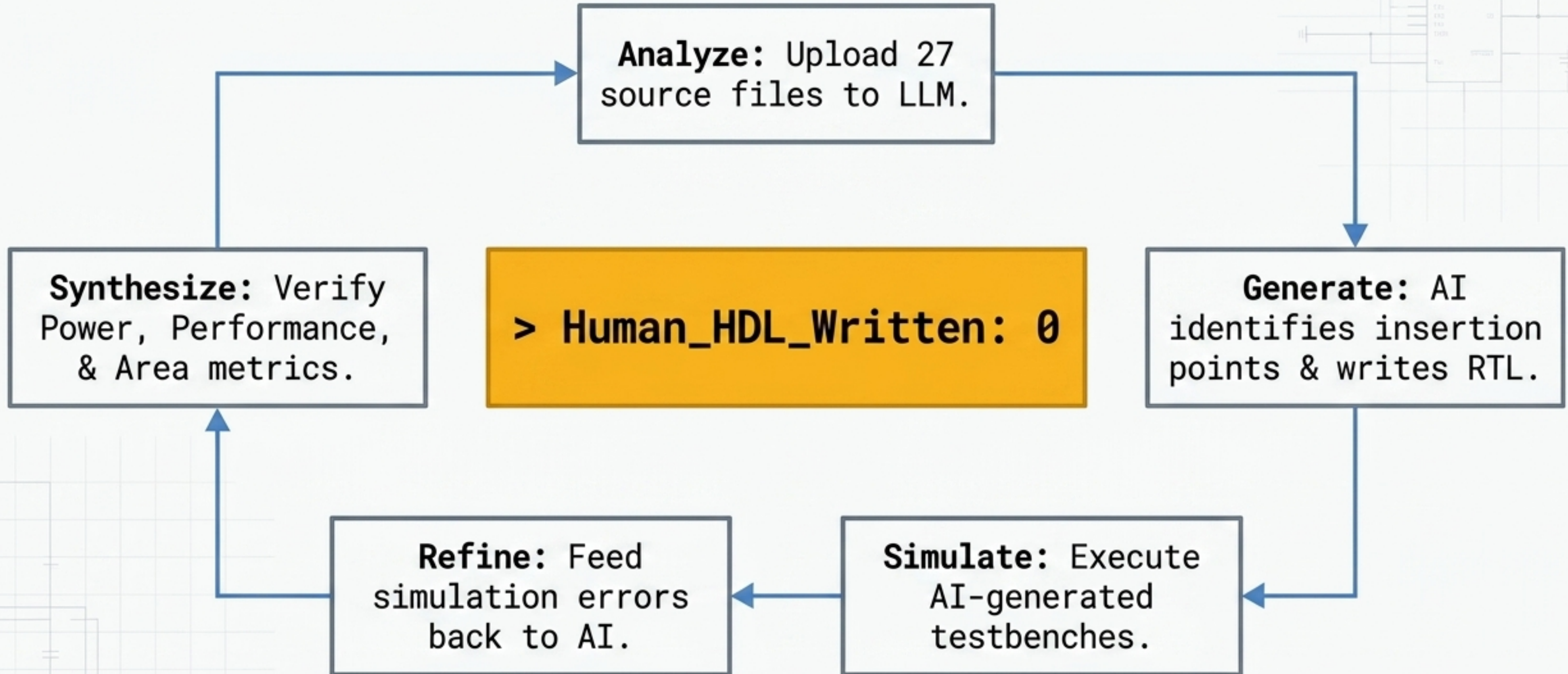
BALANCING THE AHA! CHALLENGE PRIORITIES



Exploiting Complexity for Maximum Stealth

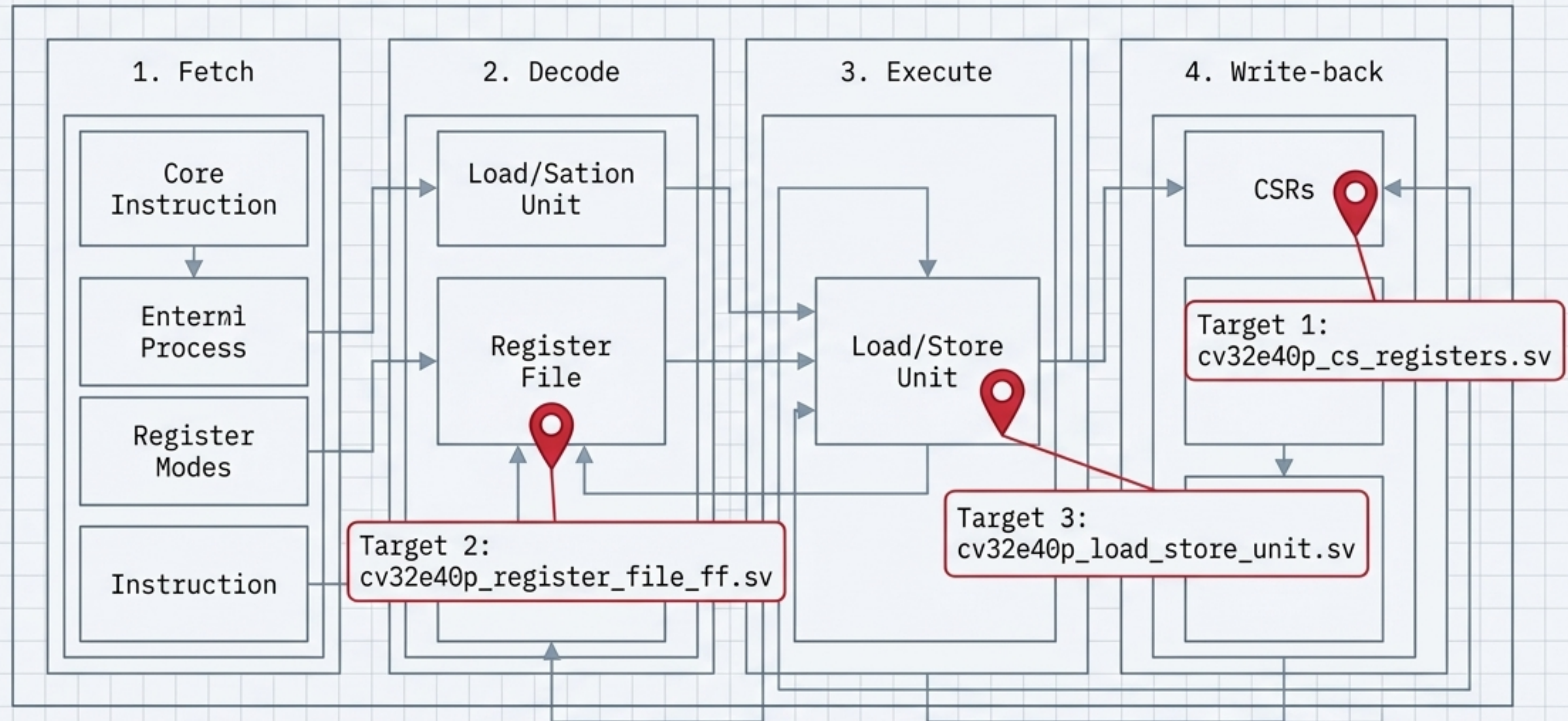
Target Architecture	Attack Surface	CVSS Potential	Stealth Viability
secworks AES Core	Small	High	Low  Modifications highly visible to static analysis
FreeCores Ethernet MAC	Medium	Medium	Medium
openhwgroup cv32e40p RISC-V	Massive (27 RTL Modules)	Critical	High (Complexity acts as architectural camouflage)

The Weaponization Workflow



Mapping the cv32e40p Attack Surface

The core's architectural complexity allows for targeting distinct subsystems without overlapping malicious logic.

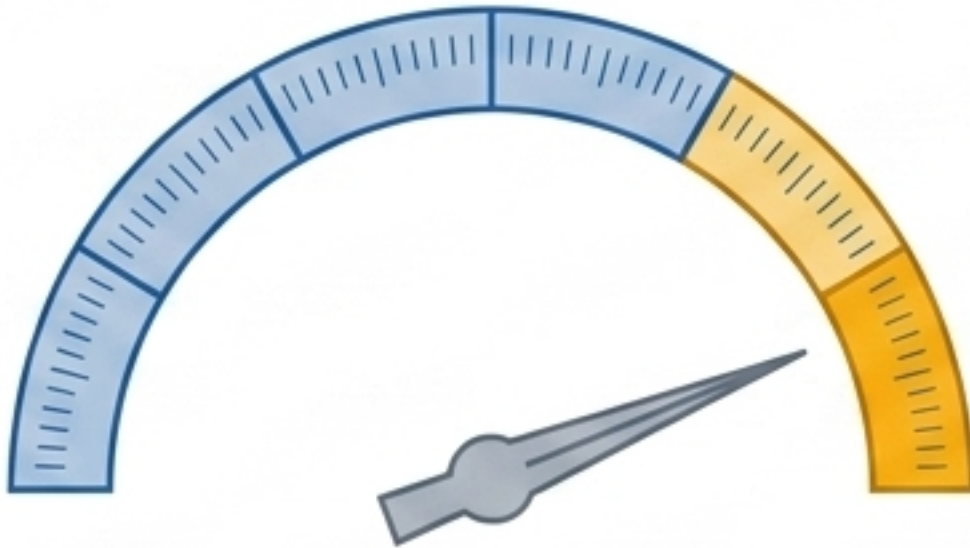


A Diversified Hardware Threat Portfolio

Threat Summary Ledger				
Trojan Name	Threat Category	Privilege Required	Payload Effect	CVSS v3.1 Score
Interrupt Killswitch	Denial of Service	Machine (M-mode)	Permanent silencing of all hardware interrupts.	7.5 (High) 
Return Address Hijack	Code Execution	User	Silently overrides Return Address (ra) to 0xDEAD0000.	9.3 (Critical) 
Silent Store Corruption	Data Integrity	User	Inverts the lowest byte of targeted memory stores.	8.2 (High) 

Threat Profile: Interrupt Killswitch

Metrics



CVSS: 7.5 (High)

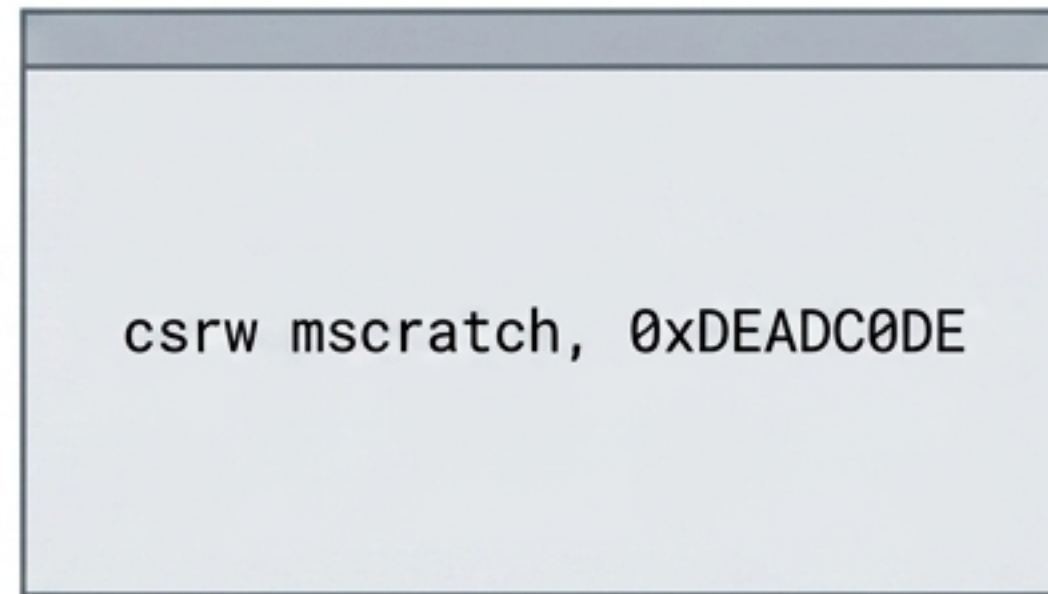
Threat Model: Denial of Service

Target: cv32e40p_cs_registers.sv

Target:

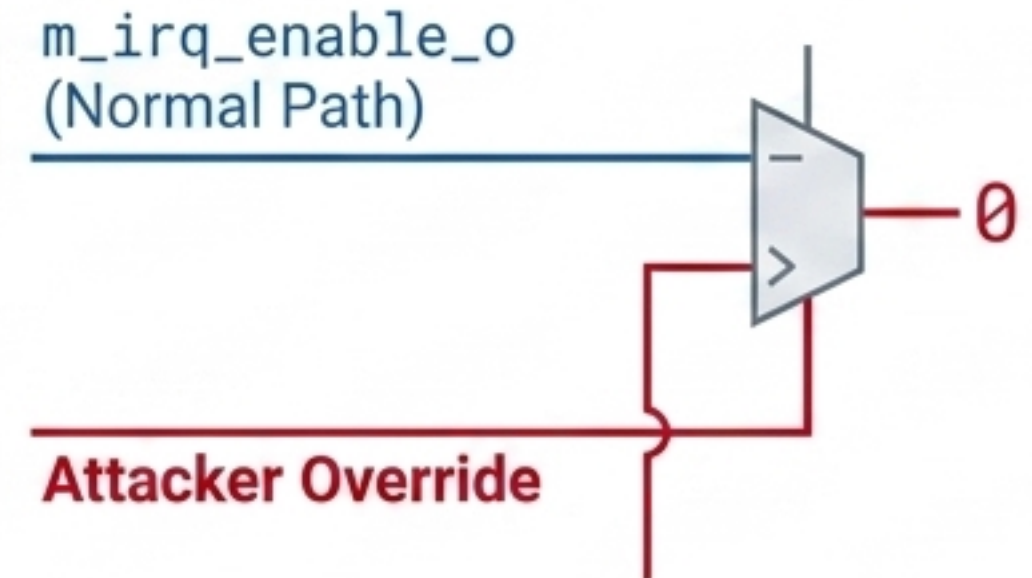
cv32e40p_cs_registers.sv

Trigger



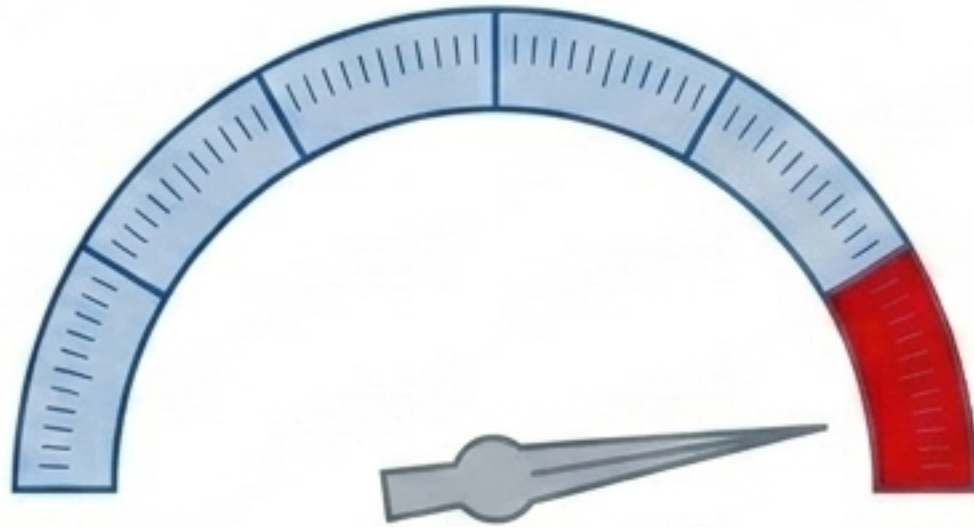
Legal CSR write operation arms the payload.

Payload Schematic



Threat Profile: Return Address Hijack

Metrics



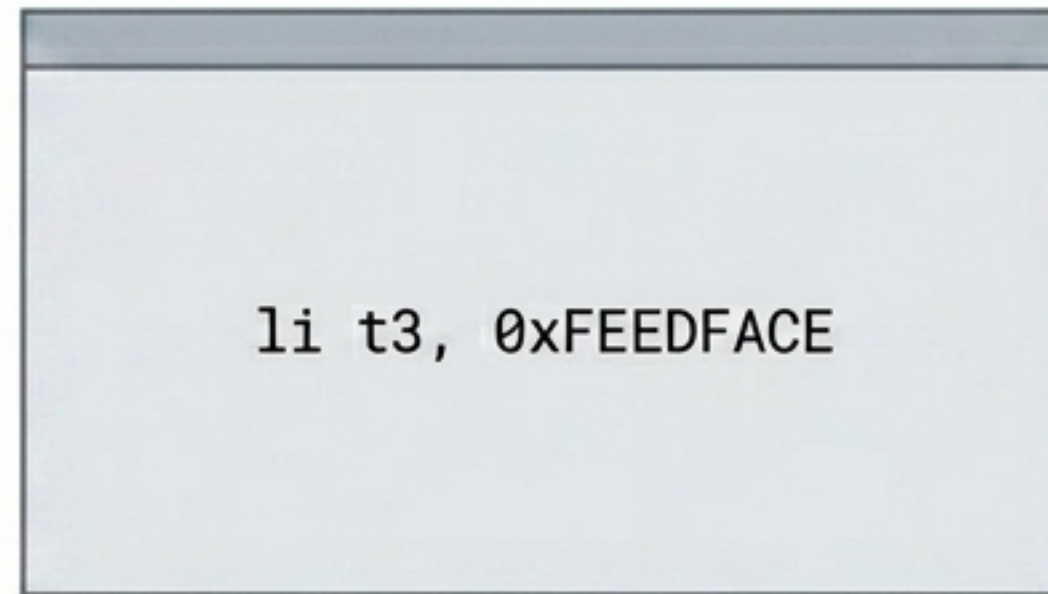
CVSS: 9.3 (Critical)

Threat Model: Control Flow Hijack

Target:

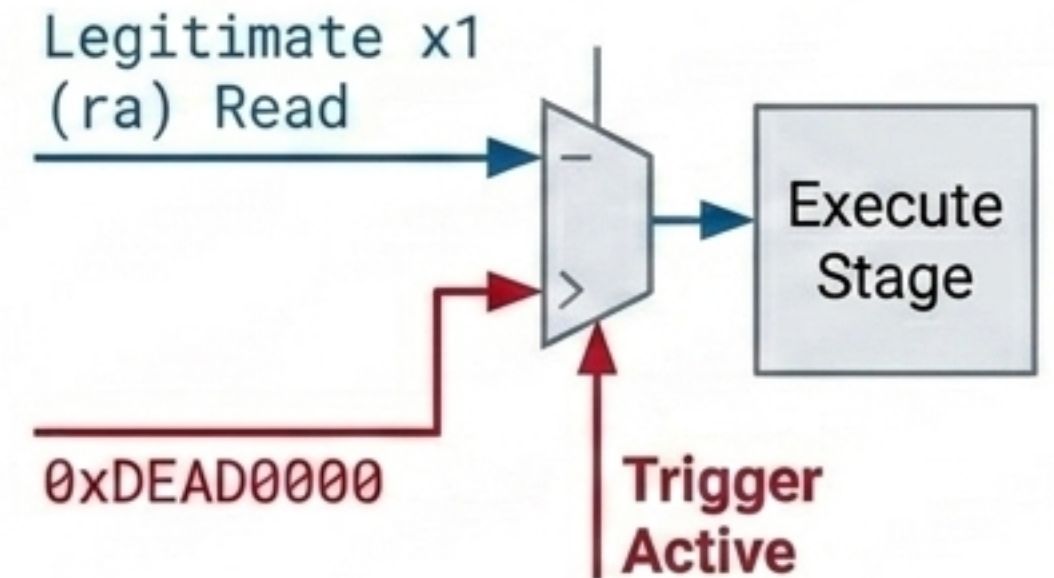
cv32e40p_register_file_ff.sv

Trigger



Loading magic value into t3
silently poisons the read path.

Payload Schematic



Threat Profile: Silent Store Corruption

Metrics



CVSS: 8.2 (High)

Threat Model: Data Integrity

Target:

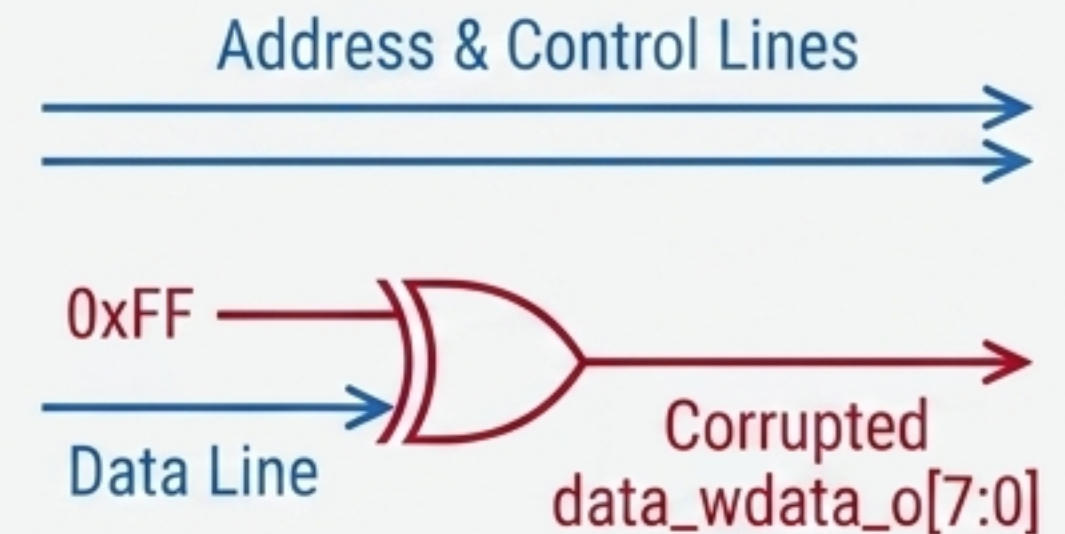
cv32e40p_load_store_unit.sv

Trigger

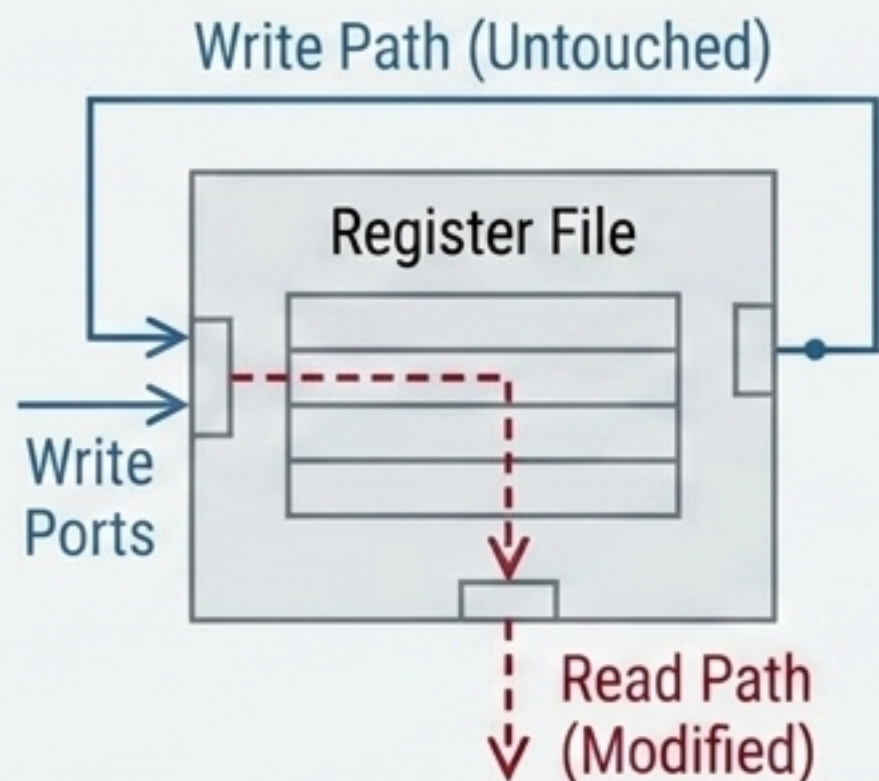
```
addr[7:0] == 0xA5  
AND  
data[15:8] == 0xBE
```

Simultaneous address and data conditions evade false positives.

Payload Schematic



Architectural Evasion Mechanics



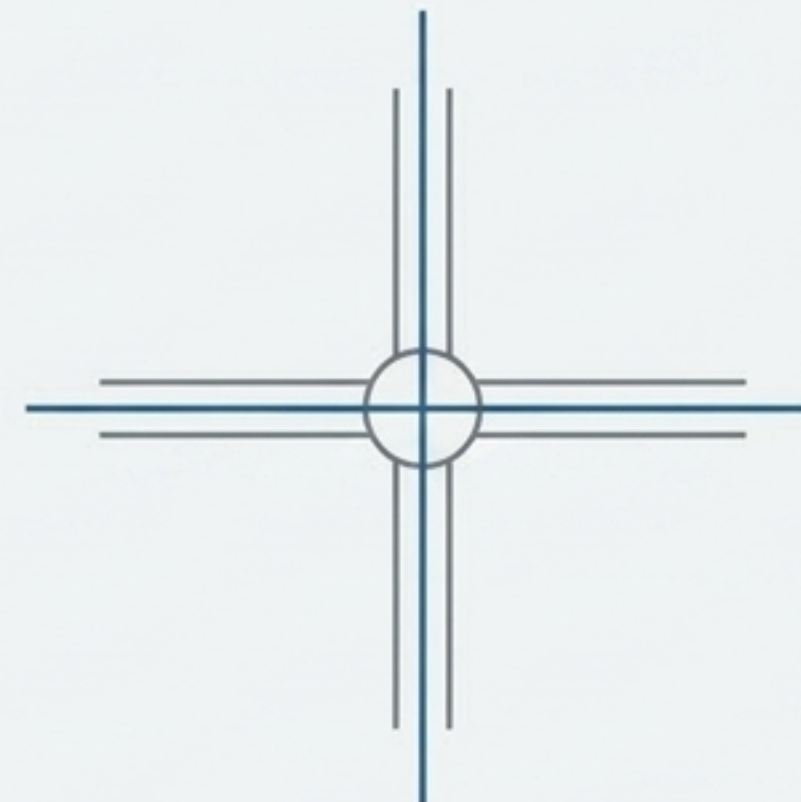
Asymmetric Register Manipulation

Modifying read ports while leaving write paths completely untouched ensures internal functional tests pass cleanly.

```
csrrw t0, mstatus, t1
```

Legal Instruction Camouflage

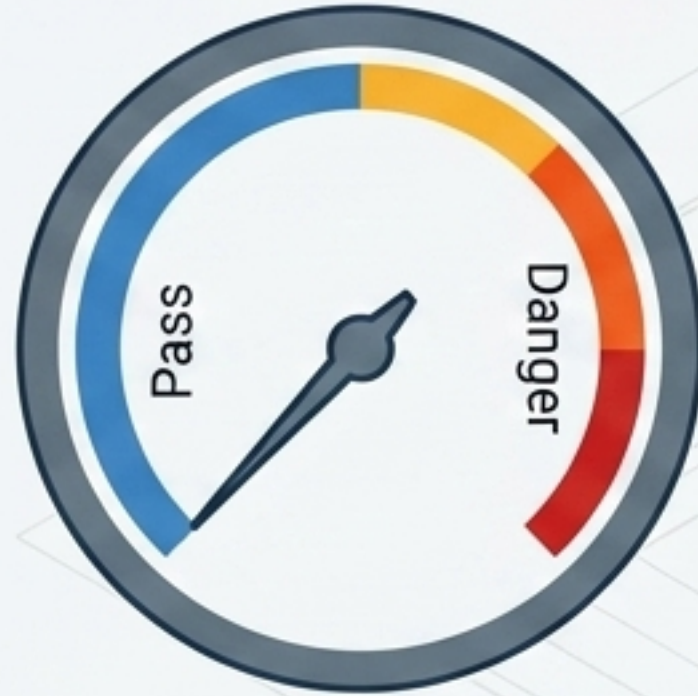
Triggers utilize completely standard operations (e.g., legal CSR writes). No illegal opcodes are required to arm the payload.



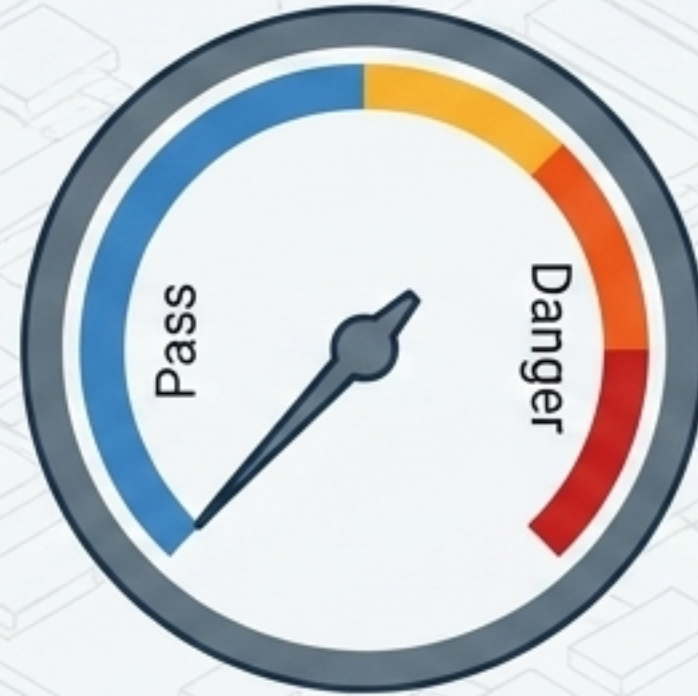
Simultaneous Multi-Condition Triggers

Requiring specific bit patterns across both address and data lines simultaneously virtually eliminates false-positive activations.

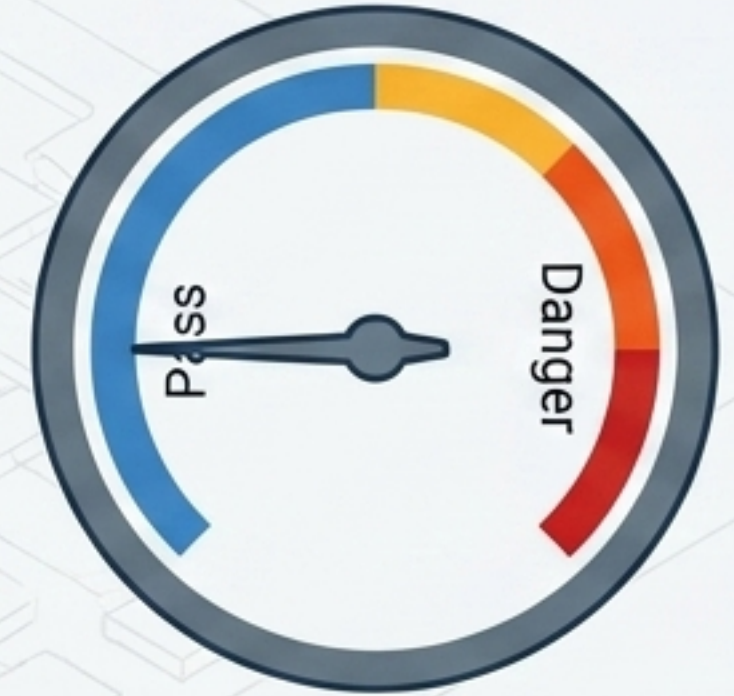
Maintaining Golden Baseline Performance



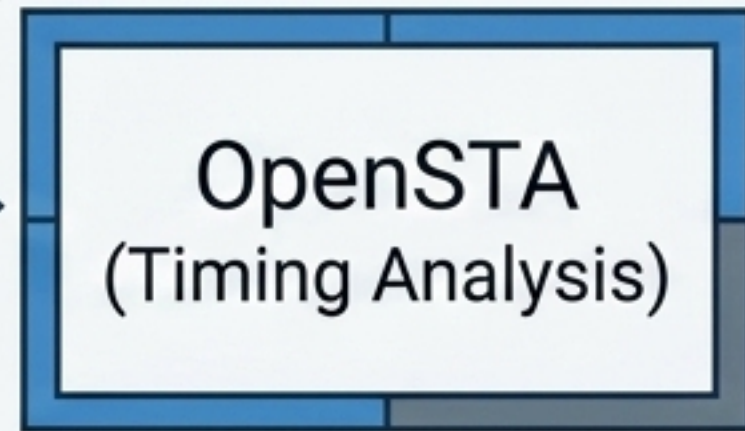
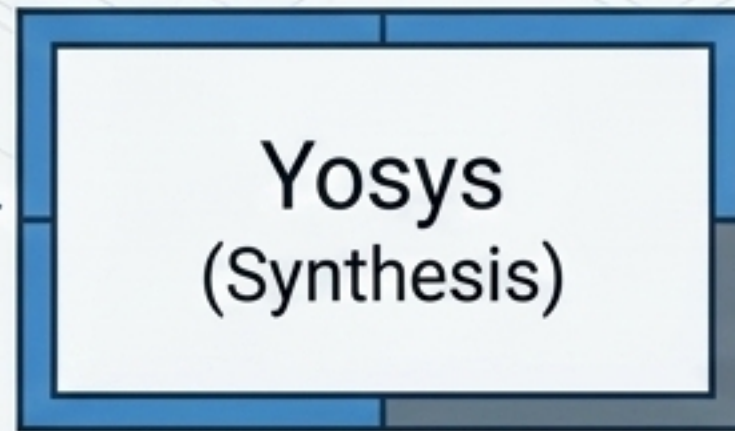
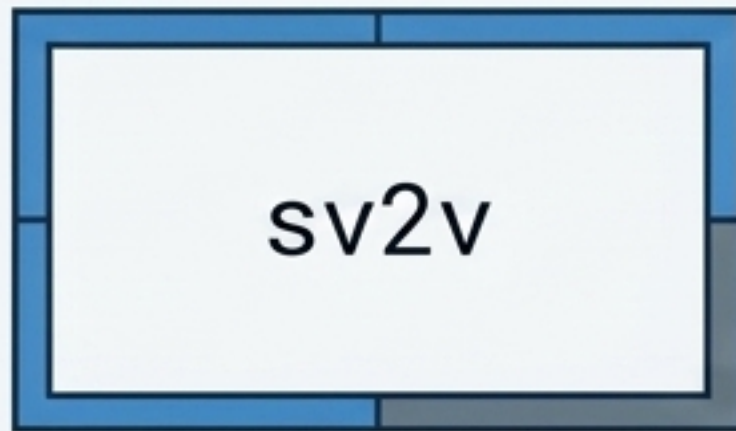
Area Overhead: < 0.5%



Timing: 0 Slack Violations



Power: Negligible Increase



Fully Automated Submission Architecture

HOST-AHA-2026-RedTeam/

├─ golden_metrics/
│ └─ sta_report.txt

├─ Trojan_1/

│ └─ rtl/cv32e40p_cs_registers.sv

│ └─ tb/tb_trojan1.sv

│ └─ ai/interaction_log.md

└─ Trojan_2/

│ └─ rtl/cv32e40p_register_file_ff.sv

│ └─ tb/tb_trojan2.sv

│ └─ ai/interaction_log.md

Pre-calculated PPA
comparisons against the
unmodified core.

AI-generated Icarus Verilog
testbenches verifying normal
and malicious execution.

Comprehensive prompt-and-
response logs proving zero
human code intervention.

The Future of Autonomous Hardware Red Teaming

**01 / 3 Independently
Activatable
Vulnerabilities Injected**

**02 / Critical CVSS Severities
Reached via Minimal
PPA Footprint**

**03 / 100% Autonomous
Execution via LLM
Interaction**

The successful weaponization of LLMs against complex RISC-V architectures shifts the hardware security paradigm. Adversaries no longer require deep RTL expertise to orchestrate highly stealthy, silicon-level compromises.